APEX RETAIL GROUP | INDEPENDENT HOMEWORK ASSIGNMENT

DOCUMENT: CLASS HOMEWORK (SESSION 8)

SCENARIO: SUBSCRIPTION CHURN & REVENUE LEAKAGE AUDIT

ASSIGNMENT OVERVIEW: You are assisting the Chief Revenue Officer at Apex Retail Group. The company's digital subscription division is experiencing unexpected revenue attrition.

You are provided two raw export files: subscription billing logs (containing dirty text, missing values, and duplicate records) and plan tier metadata. You must perform an independent EDA, clean the files, merge them, identify churn drivers, and write an executive briefing memo.

RAW DATASETS:

```Python
import pandas as pd
import numpy as np

# File 1: Subscription Logs
raw_sub_logs = {
    "sub_id": ["SUB-301", "SUB-302", "SUB-303", "SUB-303", "SUB-304", "SUB-305", "SUB-306", "SUB-307", "SUB-308"],
    "plan_code": ["PLN-1", "PLN-2", "PLN-3", "PLN-3", "PLN-1", "PLN-2", "PLN-3", "PLN-1", "PLN-2"],
    "start_date": ["2026-01-10", "2026-02-14", "2026-03-01", "2026-03-01", "2026-04-12", "2026-05-18", "2026-06-01", "2026-07-22", "2026-08-11"],
    "user_tier": [" Enterprise ", "Pro", " basic ", " basic ", "Enterprise", " PRO ", "Basic", "Enterprise", "Pro"],
    "monthly_fee": [499.00, 99.00, np.nan, np.nan, 499.00, 99.00, 29.00, np.nan, 99.00],
    "churn_status": ["Active", "Churned", "Churned", "Churned", "Active", "Active", "Churned", "Active", "Churned"]
}

# File 2: Plan Metadata
raw_plan_meta = {
    "plan_code": ["PLN-1", "PLN-2", "PLN-3"],
    "plan_name": ["Enterprise Platinum", "Pro Gold", "Basic Bronze"],
    "support_cost_per_user": [150.00, 40.00, 20.00]
}
```

PROBLEM 1: DATA CLEANING & RELATIONAL MERGING

In a Jupyter Notebook named `homework_session8_yourname.ipynb`:

1. Load both dictionaries into DataFrames `df_subs` and `df_plans`.
2. Deduplicate `df_subs`, standardize `user_tier` (strip whitespace, Title Case), convert `start_date` to datetime, and fill missing `monthly_fee` values using the median fee.
3. Left-join `df_subs` with `df_plans` on `"plan_code"`.
4. Engineer a new metric: `net_user_profit = monthly_fee - support_cost_per_user`.

PROBLEM 2: MULTI-DIMENSIONAL CHURN & PROFITABILITY EDA

1. Group the merged dataset by `churn_status` and calculate total monthly fee revenue, total net profit, and user count.
2. Group by `user_tier` and `churn_status` to determine which tier suffers the highest churn volume.
3. Group by `plan_name` and calculate average `net_user_profit` and total churn count.

PROBLEM 3: CHIEF REVENUE OFFICER BRIEFING MEMO

Write a 150-word formal executive memo to the Chief Revenue Officer detailing:

1. Total monthly revenue lost to churned subscriptions.
2. The specific user tier and plan name experiencing the worst churn rates.
3. Two actionable retention strategies backed by your EDA numbers.

SUBMISSION DEADLINE: Prior to Session 9.